PLSC 48110/28110

Lecture 1: Course introduction

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Overview

This course:

- Objectives.
- Course structure.
- Final project.

Link to syllabus.

How do we think about how experiments contribute to (social) scientific knowledge?

Reading papers

What to get out of reading a research paper:

- What is the main question of the paper?
- What method do the authors use to address the question? For empirical papers:
  - Data (Where does it come from/how is it generated? What is the sample population? What is being measured?)
  - Research design/strategy
  - Statistical tools
- What is the answer that the authors get to the main question?

Proposing theories [KKV]

(King et al., 2021)

- Propose theories that are *falsifiable*.
- If you want to be able to test this theory, it should have *observable implications*. (The more the better.)
- Clarity and precision in your theory is helpful.

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How does this inform how we think about data?

- Data is not something that is handed down to us in its complete and final form–we select and shape our data.

Principles of data collection [KKV]

1. Record and report data generation process.
2. Collect data on as many observable implications as possible.
3. Maximize the validity of measurements.
4. Ensure that data-collection methods are reliable.
5. Data and analyses should be as replicable as possible.

Observable implications

- What are these?
- How do observable implications relate to causal inference?

Inference

- What is descriptive inference?
- What is causal inference?

Elements of (descriptive/causal) inference [Holland]

(Holland, 1986)

- Define a population of interest. U
- Determine a variable that is defined over this population, for which there is variation. A
- Define response variable "of interest." Y

We can consider how A and Y vary together. Without further assumptions, this is just correlation.

What does it mean for something to cause something else?

What does it mean for something to cause something else? [Rubin]

(Rubin, 1974, 1977, 1978, 1980; Holland and Rubin, 1980)

- Possibility for multiple values of treatment–*for each individual*. Treatment *could have been different*. $S(u) = t$ OR $S(u) = c$
- Some temporal element: outcome variable is realized after "exposure" to treatment variable. $S(u) \rightarrow Y(u)$
- Conceptual possibility for multiple different versions of response variable–*for each individual*. $Y_t(u)$ may not equal $Y_c(u)$

Consider the individual *causal effect*

$$Y_t(u) - Y_c(u).$$

No causation without manipulation [Holland]

- Causes are only things that could (hypothetically) be treatments in experiments.
- NOT attributes–e.g., race and biological sex can't plausibly be manipulated, because there is no meaningful *counterfactual*.
  - What we can manipulate: others' *perception* of race/sex (e.g., audit experiments)
  - To the extent that sex/gender are social constructs, can we consider counterfactual socialization?

Fundamental problem of causal inference

- We only see the response variable under one version of treatment.
- Why does this matter?

Resolution (?) of the fundamental problem of causal inference

- Causal inference is not impossible.
- But making causal inferences without making assumptions IS impossible.
- So we need to depend on some assumptions.
- The crux: how plausible are these assumptions?

Some special cases of causal inference

- Temporal stability
- Causal transience
- Unit homogeneity
- Constant causal effect
- Independence

Temporal stability and causal transience

- The value of response will not change based on *when* you apply treatment to an observation.
- The value of response will not change if you had at some point previously applied a different treatment to an observation.

Unit homogeneity

- The value of response under a given treatment is the same for two observations; and the value of response for a different treatment is also the same.
- You can observe response on the comparable units under one of each version of treatment.

Constant causal effect

- The difference between the value of the response variable under one version of treatment as compared to another version of treatment is the same for *every* observation.
- How does this relate to the unit homogeneity assumption?
- Does this allow us to back out the value of the causal effect, without other assumptions?

Independence

- There are a "large" number of observations.
- On average, the units that are assigned one version of treatment look like the units that receive another version of treatment.

The special role of experiments

- In experiments, treatment is randomly assigned by the researcher.
- So we know independence holds by design.
- We can compare groups and get the *average* causal effect.
- (When can we also get the *individual* causal effect?)

An aside on Nobels



Figure: Esther Duflo, Michael Kremer, and Abhijit Banerjee

An aside on Nobels

2019 economics nobel awarded for: "their experimental approach to alleviating global poverty." Why is this a big deal?

- One way to understand human behavior: posit a theory. E.g., propose a utility function.

→ Deduce implications of theory.

→ (maybe) Test implications against data.

→ Update and revise theory.

Experiments

How do we know if policies are working?

- Duflo, Banerjee, Kremer motivated by work of development agencies.
- Interventions are large scale, complicated, and hard to deduce how "effective" they should be based on theory alone.
- A solution: build rigorous evidence for policy efficacy.
- One way to build evidence: experiments.

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